TEJASH RAI

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT) Bhubaneswar, Odisha, India

Bachelor of Technology in Computer Science and Engineering *Aug. 2024 -- May 2028*

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Management Systems, Computer Architecture, Systems Programming, Internet Technology

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL

ML/DL Frameworks: TensorFlow, LSTMs, Transformers, Scikit-learn, Matplotlib

LLMs & Agentic AI: LLMs, RAG, MCP (Model Context Protocol), Ollama, Faster-Whisper, Prompt Engineering

MLOps & Deployment: Docker, FastAPI, GitHub Actions (CI/CD), Azure, REST APIs

Tools & Platforms: Git/GitHub, VS Code, Google Colab, Google Cloud Platform

PROJECTS

AI-Based Methane Gas Forecasting System | *Python, LSTMs, Transformers, FastAPI, Docker* **March 2026**

- Architected a hybrid Deep Learning model utilizing LSTMs and Transformers to predict methane concentration levels, achieving high accuracy in forecasting potential leakages from multi-variate time-series sensor data.
- Engineered an end-to-end MLOps pipeline with TensorFlow for model training, and containerized the application with Docker to serve real-time inference via FastAPI, ensuring sub-second safety alerts.
- Developed a comprehensive monitoring dashboard, enabling stakeholders to visualize historical trends and receive automated push notifications for predicted anomalies.

DocPilot AI – Enterprise Document-Intelligence Platform | *Python, RAG, pgvector, Multi-Agent Systems, Docker Compose* **2026**

- Architected a multi-agent, evidence-grounded document-intelligence platform with specialized workflows for policy comparison, contract risk analysis, and natural-language knowledge search over indexed organizational documents.
- Built a shared retrieval pipeline using OpenRouter embeddings (text-embedding-3-small) and pgvector for semantic search, grounding every AI-generated answer in cited source passages with confidence scoring, and abstaining when evidence was insufficient.
- Designed a planner-coordinated multi-agent architecture (document intelligence, retrieval, citation, compliance analysis, audit) and validated it with an automated test suite of 40 passing tests at ~99% coverage, containerized via Docker Compose.

Jarvis – Agentic OS Copilot | *Python, LLMs, MCP, Ollama, Faster-Whisper, RAG* **2026**

- Architected an autonomous, multi-modal AI agent utilizing Qwen 2.5 via Ollama and Groq for high-speed LLM inference, orchestrating complex tool-calling and RAG workflows.
- Developed an end-to-end speech-to-speech pipeline by integrating Faster-Whisper for low-latency audio transcription and Kokoro TTS for natural voice synthesis.
- Leveraged the Model Context Protocol (MCP) to dynamically connect the LLM brain to external enterprise environments (Gmail, Calendar) while maintaining a secure local filesystem sandbox for production-grade reliability.

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

E-Labs, KIIT University -- Senior Member **2025 -- Present**

- **Sandrover Project (Kshitij, IIT Kharagpur):** Led Team TRRAP in designing a 4WD rover, custom-calibrating a dual-ESC powertrain to reach the national finals.
- **IOTRONIX MVP:** Named “Most Valuable Player” for demonstrating technical excellence in hardware-software integration at the 2025 E-Labs flagship event.
- **Joffee 3.0 Star Performer:** Recognized as the top performer among peers for proficiency in Java, Data Structures, and industry-standard coding practices.